

Trigonometric functions of complementary angles

The two acute angles of a right triangle add to 90° — and that single fact swaps sine with cosine.

LESSON 25

1 × 40 MIN

GEOMETRY 8, TEMA 22

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Explain why the two acute angles of a right triangle are complementary.
- Use $\sin \alpha = \cos(90^\circ - \alpha)$ and $\tan \alpha = \cot(90^\circ - \alpha)$.
- Convert a ratio of a large acute angle into one of a small angle.
- Simplify expressions that mix an angle with its complement.

TEXTBOOK

National	Tema 22, pp. 52–53
Cambridge	Extension beyond Stage 9
Workbook	—

01

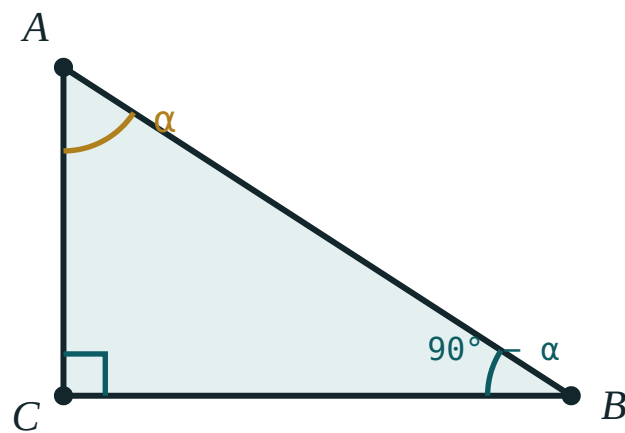
Explanation

Why the angles are complementary

In a right-angled triangle the three angles add to 180° and one of them is 90° . So the two acute angles add to 90° :

$$\alpha + \beta = 90^\circ, \text{ that is } \beta = 90^\circ - \alpha$$

Two angles adding to 90° are called **complementary**.



$$\begin{aligned}\sin \alpha &= \cos(90^\circ - \alpha) \\ \tan \alpha &= \cot(90^\circ - \alpha)\end{aligned}$$

The same two legs serve both angles — but their roles as “opposite” and “adjacent” are exchanged.

The exchange

Look at the figure. For the angle α , the leg a is opposite and b is adjacent. For the angle $\beta = 90^\circ - \alpha$, it is the other way round. So

$$\sin \alpha = \frac{a}{c} = \cos \beta = \cos(90^\circ - \alpha)$$

THE FOUR RELATIONS

$$\sin \alpha = \cos(90^\circ - \alpha) \quad \cdot \quad \cos \alpha = \sin(90^\circ - \alpha)$$

$$\tan \alpha = \cot(90^\circ - \alpha) \quad \cdot \quad \cot \alpha = \tan(90^\circ - \alpha)$$

Each function turns into its **co**-function.

That is where the names come from: “cosine” is short for *complementi sinus* — the sine of the complement.

What it is for

Old tables only listed angles up to 45° . Everything above that was read off the same page using these relations:

$$\sin 70^\circ = \cos 20^\circ \quad \cdot \quad \tan 80^\circ = \cot 10^\circ$$

Even with a calculator the relations are worth knowing, because they simplify expressions instantly:

$$\sin^2 40^\circ + \sin^2 50^\circ = \sin^2 40^\circ + \cos^2 40^\circ = 1$$

CHECK THE ANGLES REALLY ARE COMPLEMENTARY

$40^\circ + 50^\circ = 90^\circ$ ✓, so the swap is legal. For 40° and 60° it is not.

02 Worked examples

EXAMPLE 1 Write $\sin 63^\circ$ as a function of an angle below 45° .

- ① $90^\circ - 63^\circ = 27^\circ$
Find the complement.

- ② $\sin 63^\circ = \cos 27^\circ$
Sine becomes cosine.

ANSWER

$\cos 27^\circ$

EXAMPLE 2 Simplify $\sin^2 25^\circ + \sin^2 65^\circ$

① $25^\circ + 65^\circ = 90^\circ$
The angles are complementary.

② $\sin 65^\circ = \cos 25^\circ$

③ $= \sin^2 25^\circ + \cos^2 25^\circ$

④ $= 1$
The basic identity.

ANSWER**1****EXAMPLE 3** Simplify $\tan 35^\circ \cdot \tan 55^\circ$

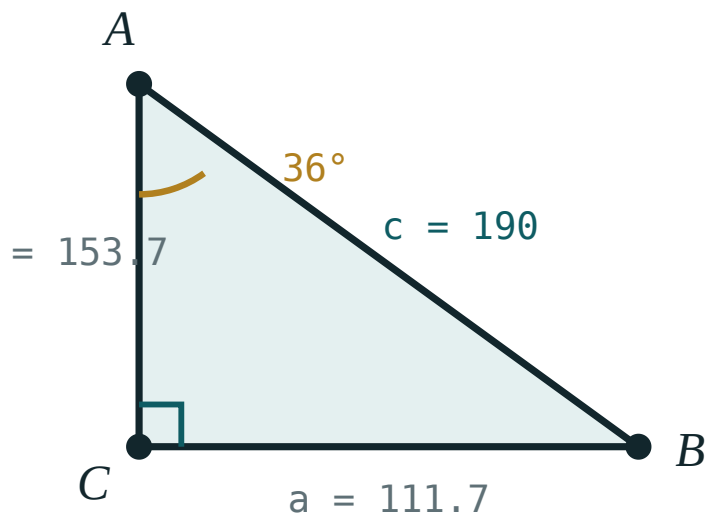
① $35^\circ + 55^\circ = 90^\circ$
Complementary.

② $\tan 55^\circ = \cot 35^\circ$

③ $\tan 35^\circ \cdot \cot 35^\circ = 1$
They are reciprocals.

ANSWER**1****03 Interactive model**

Set α to 30° and read both angles: the ratios at 30° and 60° are the same two numbers swapped.

α and $90^\circ - \alpha$ in one triangle α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Complementary angles	To'ldiruvchi burchaklar	Дополнительные углы
Sum of angles	Burchaklar yig'indisi	Сумма углов
Co-function	Ko-funksiya	Кофункция
Convert	Aylantirish	Преобразовать
Acute angle	O'tkir burchak	Острый угол
Equal ratios	Teng nisbatlar	Равные отношения
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin 70^\circ$ equals:

$\cos 70^\circ$

$\cos 20^\circ$

$\sin 20^\circ$

$\tan 70^\circ$

Two acute angles of a right triangle:

are equal

add to 90°

add to 180°

are unrelated

$\tan 35^\circ \cdot \tan 55^\circ$ equals:

0

1

2

$\tan 90^\circ$

$\sin^2 40^\circ + \sin^2 50^\circ$ equals:

1

2

0

$\sin^2 90^\circ$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write $\sin 60^\circ$ as a cosine.

ANSWER $\cos 30^\circ$

2. Write $\cos 80^\circ$ as a sine.

ANSWER $\sin 10^\circ$

3. Write $\tan 70^\circ$ as a cotangent.

ANSWER $\cot 20^\circ$

4. Write $\cot 55^\circ$ as a tangent.

ANSWER $\tan 35^\circ$

5. Find the complement of 37° .

ANSWER 53°

6. $\alpha = 28^\circ$. Find β in a right triangle.

ANSWER 62°

7. $\sin 45^\circ$ and $\cos 45^\circ$ — are they equal?

ANSWER Yes, since 45° is its own complement.

Medium · the standard question

7 PROBLEMS

1. Write $\sin 63^\circ$ as a function of an angle below 45° .

ANSWER $\cos 27^\circ$

2. Write $\cos 18^\circ$ as a sine.

ANSWER $\sin 72^\circ$

3. Simplify $\sin^2 25^\circ + \sin^2 65^\circ$

ANSWER 1

4. Simplify $\tan 35^\circ \cdot \tan 55^\circ$

ANSWER 1

5. Simplify $\frac{\sin 40^\circ}{\cos 50^\circ}$

ANSWER 1

6. Simplify $\cos^2 15^\circ + \cos^2 75^\circ$

ANSWER 1

7. In a right triangle, $\sin \alpha = 0.6$. Find $\cos \beta$.

ANSWER 0.6 — since $\beta = 90^\circ - \alpha$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\sin^2 10^\circ + \sin^2 20^\circ + \sin^2 70^\circ + \sin^2 80^\circ$

ANSWER Pair 10° with 80° and 20° with 70° : $1 + 1 = 2$

2. Simplify $\tan 10^\circ \cdot \tan 20^\circ \cdot \tan 70^\circ \cdot \tan 80^\circ$

ANSWER Pair 10° with 80° and 20° with 70° : $1 \cdot 1 = 1$

3. Simplify $\frac{\sin 32^\circ}{\cos 58^\circ} + \frac{\cos 32^\circ}{\sin 58^\circ}$

ANSWER $1 + 1 = 2$

4. Prove that $\sin \alpha \cdot \cos(90^\circ - \alpha) + \cos \alpha \cdot \sin(90^\circ - \alpha) = 1$

ANSWER Both terms become $\sin^2 \alpha$ and $\cos^2 \alpha$.

5. If $\sin(2x) = \cos(3x)$ and both angles are acute, find x .

ANSWER $2x + 3x = 90^\circ$, so $x = 18^\circ$.

6. Arrange $\sin 20^\circ$, $\sin 50^\circ$, $\cos 20^\circ$, $\cos 50^\circ$ in increasing order.

ANSWER $\sin 20^\circ < \sin 50^\circ = \cos 40^\circ < \cos 20^\circ$; and $\cos 50^\circ = \sin 40^\circ$, so the order is $\sin 20^\circ < \cos 50^\circ < \sin 50^\circ < \cos 20^\circ$.

7. Simplify $(\sin 25^\circ + \cos 65^\circ)^2$

ANSWER $\cos 65^\circ = \sin 25^\circ$, so $(2 \sin 25^\circ)^2 = 4 \sin^2 25^\circ$

07 Homework

Homework — 5 problems

Geometry 8, Tema 22, pp. 52–53.

1. Write $\sin 74^\circ$, $\cos 12^\circ$ and $\tan 66^\circ$ as functions of angles below 45° .

2. Simplify $\sin^2 31^\circ + \sin^2 59^\circ$

3. Simplify $\tan 22^\circ \cdot \tan 68^\circ$

4. Simplify $\frac{\cos 41^\circ}{\sin 49^\circ}$

5. If $\sin(3x) = \cos(x + 10^\circ)$ with both angles acute, find x .